

The empirical recurrence for OEIS A237386, recovered from its tail

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Abstract

OEIS A237386 records “Empirical recurrence of order 58” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 434$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 60$. Its last coefficient is $c_{58} = -785664 \neq 0$, so no further tail satisfies anything shorter; any order-58 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A237386 is “Number of $(n+1) \times (2+1) \dots 2$ arrays colored with the maximum plus the upper median minus the minimum of every 2×2 subblock.”. It has offset 1 and begins

434, 3240, 24340, 183764, 1379886, 10406596, 78612538, 593121116, ...

The entry states:

Empirical recurrence of order 58 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 10:08:40, revision 7) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 434$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 434$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 58: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 58.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 868$ exact terms from $n = 2$ on, it returns order 58 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{58} c_i a(n-i),$$

c_1	0	c_{16}	−43283142	c_{31}	−907186621	c_{46}	40905143707
c_2	4	c_{17}	−381003298	c_{32}	−107721043473	c_{47}	−23146895860
c_3	105	c_{18}	−135460215	c_{33}	149643760415	c_{48}	−16192681235
c_4	1317	c_{19}	−189758653	c_{34}	33030829811	c_{49}	8919000860
c_5	4402	c_{20}	−80639381	c_{35}	−214268801845	c_{50}	3218319646
c_6	15312	c_{21}	3897725084	c_{36}	111089872770	c_{51}	−3502546480
c_7	37385	c_{22}	596638781	c_{37}	81373051894	c_{52}	506770108
c_8	−61634	c_{23}	−7163891360	c_{38}	−185123862794	c_{53}	900159896
c_9	−368147	c_{24}	5010251579	c_{39}	123914654188	c_{54}	−416905796
c_{10}	−1137893	c_{25}	−2092877225	c_{40}	84417009659	c_{55}	−177042032
c_{11}	−2752433	c_{26}	−11235975741	c_{41}	−186826934701	c_{56}	80961040
c_{12}	2463344	c_{27}	31900233210	c_{42}	53106332953	c_{57}	11411904
c_{13}	15243868	c_{28}	−19118035198	c_{43}	72815246477	c_{58}	−785664
c_{14}	23241200	c_{29}	−57117879661	c_{44}	−77677361339		
c_{15}	63483255	c_{30}	84712664832	c_{45}	17043040729		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 60$, and it is the recurrence OEIS A237386 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{58} - c_1 x^{57} - \dots - c_{58}$. Its constant term is $-c_{58} = 785664$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 58 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 58, so $\deg q = 58$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 58, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 18 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 58, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -785664 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-58 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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